

SAMIUL HOSSAIN SAJAL

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EDUCATION

(Oct 2022 - Present)

M.Sc. in Physics
University of Cologne, Germany.

Primary: Condensed Matter Physics
Secondary: Computational Physics

(Jun 2019 - Nov 2021)

M.Sc. in Physics
SUST, Bangladesh.

(Mar 2015 - Apr 2019)

B.Sc. in Physics
SUST, Bangladesh

LANGUAGES

- English — Fluent (C1)
- German — A1 (currently attending a German course)
- Bangla — Native

SUMMARY

Condensed matter physics graduate student with experience in TEM image segmentation, grain tracking, and molecular dynamics simulation. Currently developing computer-vision workflows for materials analysis at Forschungszentrum Jülich (IAS-9) using PyTorch, SAM-based models, and HPC.

RESEARCH & PROFESSIONAL EXPERIENCE

Student Research Assistant (Nov 2025 - Present)
IAS-9, Forschungszentrum Jülich

- Master's thesis focused on segmentation and tracking of in-situ TEM data.
- U-Net, SAM2, SAM3 and MatSAM workflows for TEM image and video analysis
- Grain tracking using overlap analysis and Hungarian matching
- Post-processing, topology refinement and Validation against manually annotated data

Master's Thesis Research (Oct 2019 - Nov 2021)
Shahjalal University of Science & Technology (SUST)

- Molecular Dynamics simulations using ReaxFF and LAMMPS
- HPC-based analysis of aluminum nanoparticle formation and oxidation
- Temperature-dependent growth analysis with scientific visualization

SKILLS

Core Paradigms

- Computer Vision, Image & Video Segmentation, Multi-Object Tracking, Machine Learning

Models & Architectures

- SAM / SAM2 / SAM3, MatSAM, U-Net, Visual Prompt Engineering

Frameworks & Libraries

- PyTorch, OpenCV, SciPy, Scikit-Image, NumPy, Matplotlib

HPC & Compute

- Linux/Ubuntu, Multi-GPU Workflows, CUDA, Git/GitHub, LaTeX

Other Methodologies

- Molecular Dynamics (LAMMPS, ReaxFF), EBSD Analysis, Napari

SELECTED PRESENTATIONS

Poster Presentation: Atomic-scale investigation of the formation of Al nanoclusters in the Pulsed Laser Ablation Technique.

SELECTED COURSEWORK & TRAINING

Deep Learning and Physics – *University of Cologne* (Feb 2026)
Deep Learning with PyTorch: Image Segmentation – *Coursera Project* (Oct 2025)
Computational Many-Body Physics – *University of Cologne* (Feb 2025)
Scientific Programming with Python – *University of Cologne* (Jul 2023)
Google Data Analytics Professional Certificate – *Coursera* (Sep 2022)